Day 3: Networking Foundations & Wireshark Analysis

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1. Introduction

If hacking is the art of entry, networking is the map of the building. To find vulnerabilities in a system, you must first understand how data moves from Point A to Point B. Today was dedicated to stripping away the user interface and looking at the raw pulses of data **packets** that travel across a wire.

2. Objective of the Day

The goal of Day 3 is to master the fundamentals of communication protocols and learn how to use a packet sniffer (Wireshark) to "eavesdrop" on network traffic in our secure lab environment.

3. Core Networking Concepts

A. The TCP/IP Model

Everything on the internet follows this 4-layer structure:

- **Application Layer:** Where your browser (HTTP) or email (SMTP) lives.
- **Transport Layer:** Ensures data arrives correctly (TCP/UDP).
- **Internet Layer:** Routes packets using IP addresses.
- **Network Access Layer:** Handles physical hardware (Ethernet/Wi-Fi).

B. IP vs. MAC Address

- **IP Address (Logical):** Like your house address; it can change if you move (e.g., `192.168.1.10`).
- **MAC Address (Physical):** Like your Social Security Number; it is burned into your hardware at the factory.

C. Common Ports (The "Doors")

Port	Protocol	Service
21	FTP	File Transfer
22	SSH	Secure Remote Login
80	HTTP	Unsecured Web Traffic
443	HTTPS	Secured Web Traffic

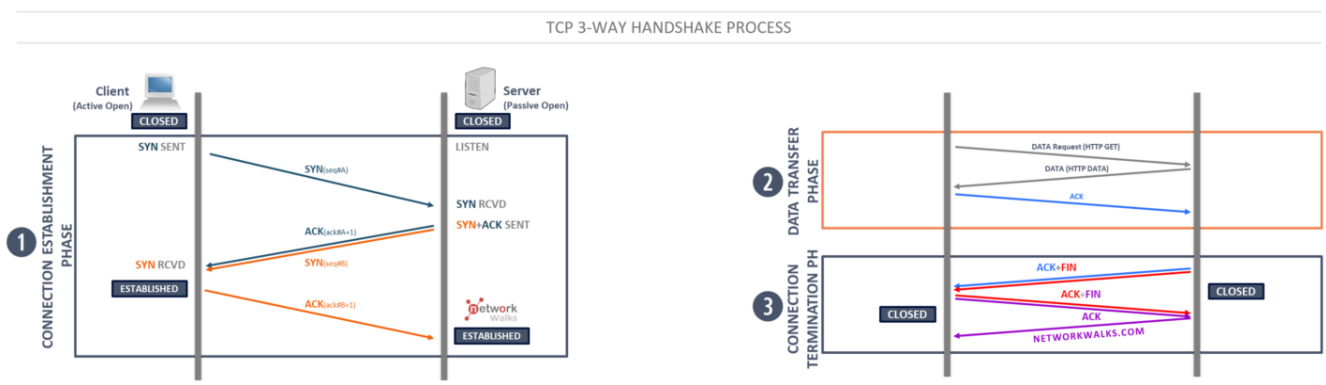
4. Tools Used: Wireshark

Wireshark is a "Packet Sniffer." It captures every piece of data entering or leaving your network card and translates it into a format humans can read.

- **Promiscuous Mode:** Allows the network card to see all traffic on the network, not just traffic intended for it.
- **Filters:** Help us find specific things (e.g., typing `http` in the bar shows only web traffic).

5. What I Did Today

1. **Protocol Analysis:** I studied the difference between **TCP** (reliable, requires a handshake) and **UDP** (fast, but no confirmation of receipt).
2. **The Three-Way Handshake:** I used Wireshark to observe the **SYN -> SYN-ACK -> ACK** process that happens every time a connection is made.
3. **Lab Traffic Capture:** I started a capture in Kali, then pinged my Metasploitable machine. I watched the **ICMP** (Ping) packets appear in real-time.
4. **Network Tools:** I practiced using `netstat -rn` to see my routing table and `nslookup` to see how domain names turn into IPs.



6. What I Learned

- **Everything is Visible:** I realized that on an unencrypted connection (like HTTP or FTP), passwords can be seen in plain text through Wireshark.
- **Protocol Hierarchy:** Understanding that a packet is like an envelope inside an envelope; each layer adds its own "header" info.
- **Troubleshooting:** I learned that `traceroute` is vital for seeing where a connection is failing between multiple routers.

Key Takeaway for Newbies:

"Networking isn't just about connecting computers; it's about understanding the language they speak. If you know the language, you can hear the secrets."

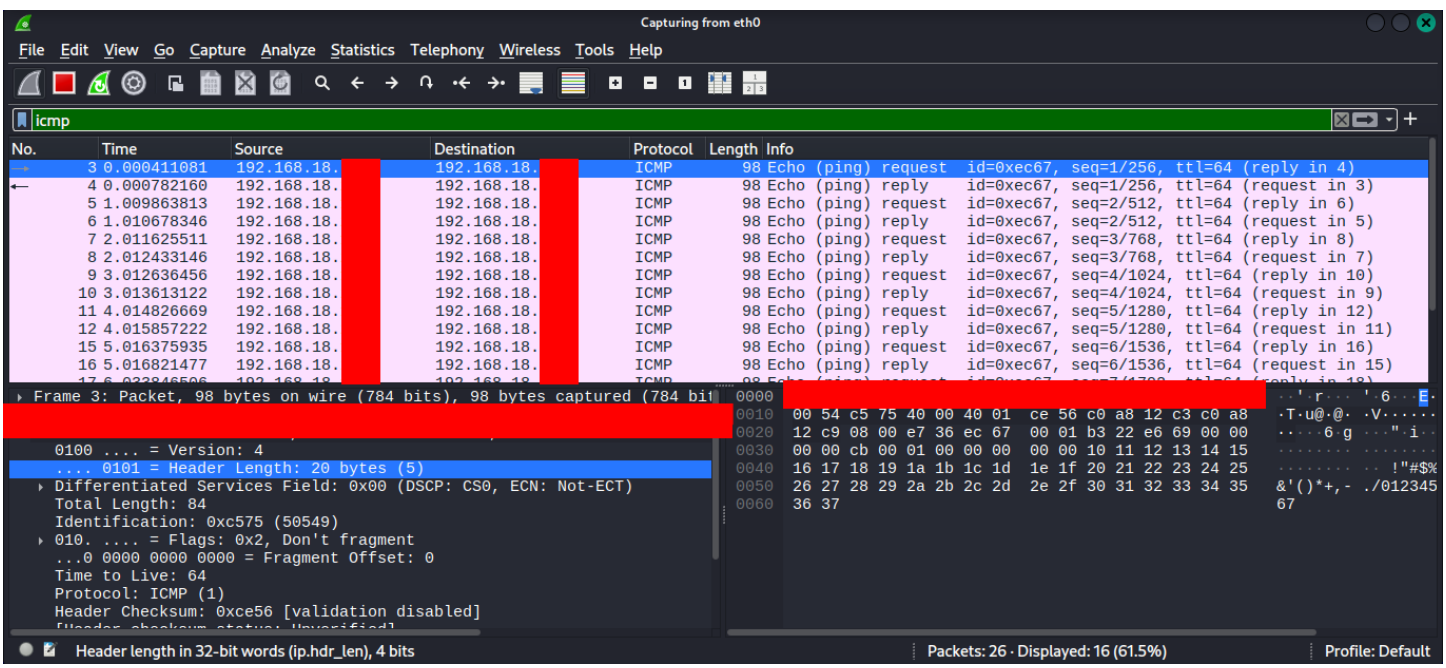
7. Hands-on Examples & Practice

- **Task:** Identify the IP of your lab gateway and capture a "Ping" request in Wireshark.

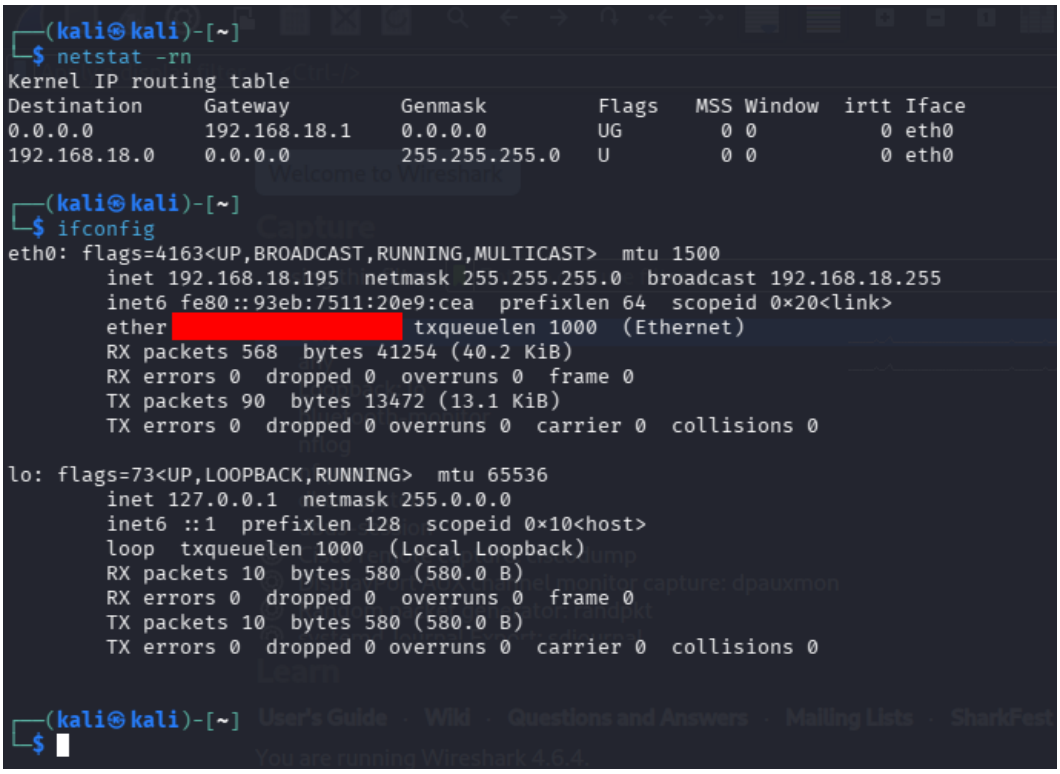
Execution:

1. Open Wireshark and start capturing on `eth0`.
2. Open Terminal and type: `ping -c 4 [Metasploitable-IP]`.
3. Go back to Wireshark and type `icmp` in the filter bar.
4. Observe the "Request" and "Reply" packets.

Screenshot 1: Wireshark running with the `icmp` filter active, showing the ping packets.



Screenshot: A terminal window showing the output of `netstat` or `ifconfig`.



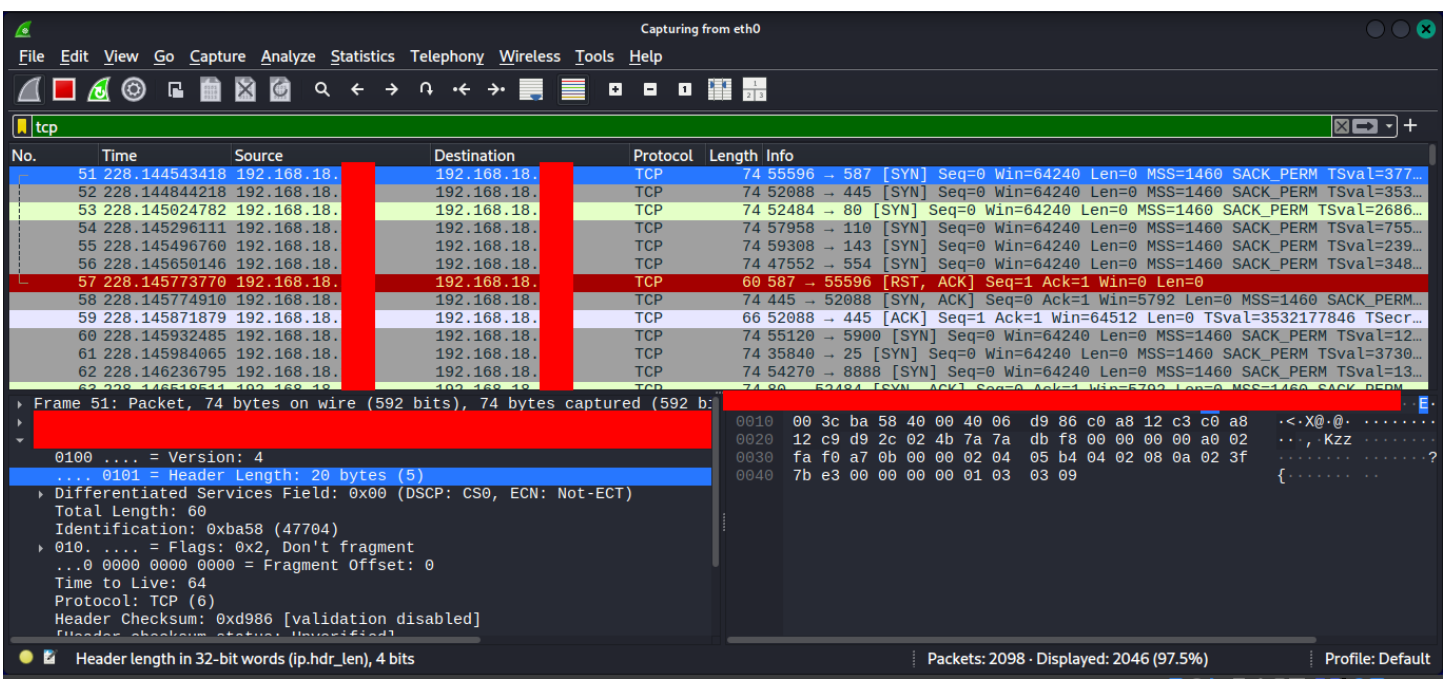
```
┌──(kali㉿kali)-[~]
└─$ netstat -rn
Kernel IP routing table
Destination     Gateway         Genmask         Flags   MSS Window  irtt Iface
0.0.0.0         192.168.18.1    0.0.0.0         UG        0 0          0 eth0
192.168.18.0    0.0.0.0         255.255.255.0   U         0 0          0 eth0

┌──(kali㉿kali)-[~]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.18.195  netmask 255.255.255.0  broadcast 192.168.18.255
        inet6 fe80::93eb:7511:20e9:cea  prefixlen 64  scopeid 0x20<link>
        ether [redacted]  txqueuelen 1000  (Ethernet)
        RX packets 568  bytes 41254 (40.2 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 90  bytes 13472 (13.1 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 10  bytes 580 (580.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 10  bytes 580 (580.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

┌──(kali㉿kali)-[~]
└─$
```

Screenshot 3: A close-up of a TCP packet in Wireshark showing the "Source" and "Destination" ports.



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